WCC FORGE HACKATHON / ROUND 1

RelayMap

Evidence-First Project Continuity and Knowledge Handover Platform

CORE PROMISE

When a contributor moves on, RelayMap gives the successor a sourced view of the project history, outstanding questions, possible knock-on effects and practical next steps, within the access they have been granted.

Designed initially for student project groups and compact software teams, RelayMap brings together useful context from shared files, work items and approved repository records.

This Round 1 proposal uses the supplied seven-page Hackathon Challenge Brief as its reference.

PROPOSAL STATUS

The pages that follow set out the intended product, operating process, system design, safeguards, delivery boundaries and evaluation approach. They describe a proposal; they do not represent completed implementation or test results.

01 / OVERVIEW

Understanding of the given problem statement

The complete proposal in one view.

When a contributor exits a team, the remaining members can inherit the files without inheriting the understanding needed to use them. Decisions may be spread across meeting records, work trackers, repositories and conversations, while practical details remain with individual contributors. A new owner may locate the code and documentation yet still be unsure why an approach was selected, which pieces are unfinished, what relies on them, or who can help remove a blocker. RelayMap is intended to reduce this gap during the transition.

The platform assembles permitted records from uploaded files and one focused live connector: GitHub, chosen for the first software-team use case. It converts the collected material into a shared set of project records, connects related work and decisions, checks for transition risks, and prepares a handover tailored to its recipient. Claims in that handover retain links to their supporting material. Where the available record is incomplete, outdated, inconsistent or only suggestive, RelayMap says so rather than presenting an assumption as fact. An authorised reviewer must approve the recipient-specific version before distribution.

THE DISTINCTIVE IDEA

Instead of stopping at a recap of past activity, the Continuity Intelligence Engine examines which duties may lose an owner and which connected work may be affected by the person's departure.

At a glance	Proposed decision
Initial users	Student groups and small teams building software or digital products.
Primary transition	One contributor exits and a successor takes over, or the remaining team shares the responsibilities.
Initial sources	Uploaded PDF, DOCX, Markdown and CSV files, together with a single GitHub connection restricted to read access.
Core outcome	A reviewer-approved guide for the recipient, including source references, unassigned work, task relationships and the first actions to take.
AI approach	RAG-assisted retrieval and schema-based extraction, followed by rule checks and review by an authorised person.
Initial stack	React for the interface; FastAPI for services; PostgreSQL and pgvector for project data and retrieval; document parsers and an LLM adapter that is not tied to one vendor.

Scope boundary: the first release is designed to complete one dependable transition workflow. RelayMap is not intended to become another project-management suite, connect to every enterprise service at once, or decide staffing and task ownership without human approval.

02 / PROBLEM DEFINITION

Users affected by the problem

A manageable first audience, with room to expand once the workflow is proven.

The supplied brief points to a gap between having project artefacts and being able to act on them. The files may remain available, but decision rationale, incomplete activities, inter-task dependencies, recurring trouble and useful contacts can be distributed across several places. A departing contributor takes part of the explanation with them. RelayMap concentrates on the handover period, when this missing context can interrupt otherwise continuing work.

Initial target environment

The initial users will be student project groups and smaller software teams. Their work commonly leaves behind observable records such as issue trackers, pull requests, meeting notes, technical guides and task exports. The team can also reach these users directly for interviews, while a demonstration can rely on synthetic or explicitly permitted material. The same design could later serve research groups, volunteer organisations and workplace teams.

Primary users

User	Need from RelayMap
Departing member	A guided way to add missing know-how and check which duties are moving to someone else.
Project lead / authorised reviewer	One place to inspect project status, source material, transition gaps, risks and pending approvals.
Incoming member	A handover relevant to their role, showing assigned work, key decisions, linked tasks, blockers and immediate next steps.
Workspace administrator	Tools to manage integrations, account links, access policies, data retention and removal of access.

Representative scenario

Consider a student group of four building an AI application. Its backend contributor is about to leave. The team's GitHub repository contains active issues and pull requests, meeting notes describe the reasoning behind the API design, a README lists local setup steps, and a spreadsheet records task assignments. The successor can inspect the code but cannot easily tell which API items remain open, what prompted a change to authentication, which frontend activities are waiting on the API, or which setup instructions exist only in the outgoing developer's memory. RelayMap brings these records into one view, checks work linked to that contributor, surfaces dependent activities, asks focused follow-up questions and prepares a sourced draft for approval.

THE PROBLEM WE SOLVE

Help a team retain usable knowledge through a specific role change. RelayMap is not a productivity surveillance product or a substitute for a project-management system.

03 / PRODUCT DESIGN

Proposed Solution

The platform combines four connected capabilities to prepare a usable transfer of project knowledge.

1. Unified project knowledge

After permission checks, RelayMap turns material from each source into standard records. These can represent a decision, task, responsible person, dependency, blocker, unresolved question, repeated issue, useful contact or follow-up action. Each record retains where it came from, its relevant dates and the access rules inherited from the source. The data model also makes clear whether an item was directly recorded, inferred by the model or added by a team member.

2. Project Knowledge Map

The Project Knowledge Map displays links among contributors, work items, decisions, reported issues and dependent activities. This lets a successor trace how a blocked item affects later work. Relationships already stated in the source systems are stored separately from connections suggested by AI; suggested links remain unconfirmed until a person or stronger evidence validates them.

3. Continuity Gap Detector

The Continuity Gap Detector is the product's main differentiator. It looks for work that may lose a responsible owner or cause delays as a named contributor exits. It begins with records assigned to that person or to an identity verified as theirs, then checks whether the item is still open, whether someone has taken it over, and whether completion or reassignment is recorded. It also follows any known downstream dependencies. A passing reference to someone in a conversation is not enough to classify work as orphaned.

4. Evidence-Linked Handover

The resulting handover gives the incoming contributor a practical starting point: the project snapshot, decisions with an approved status, work still in progress, ownership gaps, connected tasks, risks, unanswered questions, relevant contacts and suggested first steps. Important statements point to the original source and a page, section or record identifier wherever the source provides one. When evidence is missing, RelayMap leaves the point open for confirmation instead of completing the gap with generated detail.

Capability	What makes it useful
Knowledge states	Keeps decision status separate from evidence quality, distinguishing proposals and approvals while identifying disagreement, ageing or weak support.
Exit knowledge capture	Uses the outgoing contributor's known areas of work to prepare questions about information that existing records do not contain.
Role-specific views	Builds the successor's report from information available under that person's permissions, rather than reusing an unrestricted report.
Human approval	Leaves release approval to an authorised reviewer and records edits and decisions in the handover history.

04 / CORE WORKFLOW

Complete Product Workflow

The following sequence follows the five operating stages set out in the supplied challenge brief.



A controlled workflow: no generated handover is shared before authorised review and approval.

01 Connect

A project lead opens a workspace, adds the relevant members and links each person to verified platform accounts. The lead can add PDF, DOCX, Markdown and CSV material and connect one chosen GitHub repository through a repository-limited, read-only permission. Before any content enters the processing pipeline, the source is given an access classification. Anything not classified is denied by default.

02 Reconstruct

In this stage, the imported material is parsed into a consistent format. The system identifies people, tasks, decisions and other project entities, then creates a dated sequence and a map of their relationships. Status and dependency information explicitly supplied by a source system takes precedence. A relationship proposed by the model is labelled as an inference until someone verifies it.

03 Verify

Each significant statement is checked against its supporting record. RelayMap confirms that the referenced source and location can be found and that the intended recipient is allowed to open it. It also marks disagreement between sources, absent support, older information and incomplete sync runs. The newest entry does not automatically win; the system considers which source is authoritative for that type of fact and whether a decision was formally approved.

04 Review

Before release, a person with the required authority checks the draft alongside its evidence excerpts, uncertainty labels and flagged continuity issues. They may revise a statement, add missing context, reject an extraction, remove sensitive material or approve an individual item and the complete report. RelayMap records the reviewer, the change and its time so the decision can be audited later.

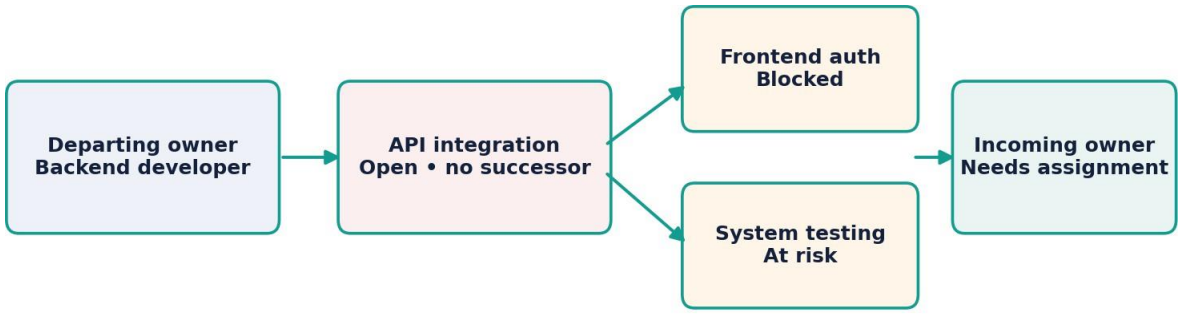
05 Handover

Once approved, RelayMap prepares the version intended for the successor using that person's identity and effective permissions. The handover summarises the known project position, responsibilities being transferred, unfinished items, dependencies, risks, relevant decisions and an ordered list of initial actions. The recipient may confirm that the handover was received. That acknowledgement records receipt only; it is not a certification that every entry is complete or accurate.

05 / MAIN FEATURES

Continuity Intelligence Engine

Surface work that may lose an owner and the activities that could be held up by the transition.



Illustrative dependency chain. Arrows must be source-backed or explicitly labelled as inferred.

Identity resolution

One contributor may appear under several usernames or account IDs across the tools a team uses. During setup, the administrator associates accounts that have been confirmed as belonging to the same person with one RelayMap user record. Similarity-based matching may help find candidates, but it cannot settle an uncertain match by itself. A name appearing in a comment is not accepted as evidence that the person owns the task.

Orphaned-work candidate rule

1. **Assignment match: an authoritative record assigns the work to the departing person or to an account that has been verified as theirs.**
2. **Work remains open: the latest trusted status available does not mark the item complete, cancelled or closed.**
3. **No replacement recorded: RelayMap cannot find a confirmed transfer to another owner.**
4. **Source check: the underlying record can be accessed and is recent enough to assess; conflicts and incomplete retrieval are surfaced.**
5. **Dependency check: known downstream work is displayed. Any relationship inferred by the model remains visibly unverified until reviewed.**

IMPORTANT CLASSIFICATION RULE

Matching these checks makes an item a possible ownership gap; it does not establish that the task has been abandoned. A reviewer must confirm, dismiss or arrange a transfer. Where the owner or status cannot be established, the item remains marked “needs verification”.

Explainable risk prioritisation

The first risk indicator is meant to help reviewers decide what to inspect first; it is not a forecast of whether the project will fail. Its inputs are visible: whether work is still open, whether a successor is missing, how many important items depend on it, whether it is blocked, how old its evidence is and whether the source record is disputed. RelayMap displays the reasons behind each priority level. Thresholds can be tuned against labelled examples, but the score is not presented as a scientifically validated probability.

05 / MAIN FEATURES

Evidence, time and project-state model

RAG helps find relevant passages; structured project records and explicit rules assemble the state of the work.

Retrieval-Augmented Generation is one part of the solution, not the complete reconstruction method. RelayMap uses retrieval to bring relevant passages into context, then applies structured extraction and validation to turn those passages into defined record types and links. Keeping source material, extracted records and generated explanations separate makes it possible to inspect the reasoning and prevents a generated summary from becoming the project's authoritative record.

Core record types

Record	Key fields
Source	Workspace identifier, connector or file type, source-system identifier or URL, author, creation and update dates, import run, content fingerprint and permissions.
Knowledge claim	The statement, its category and related project entity, decision/review state, evidence condition, event date, extraction date and reviewer.
Evidence link	Originating record, page or section or issue/comment reference, supporting excerpt, source revision and access restriction.
Task / responsibility	Persistent work-item ID, verified owner, current status, any supplied due date, originating platform and successor or reassignment details.
Dependency edge	The two connected records, the link type, whether the link came from a source or model inference, its evidence and confirmation status.
Handover / review	Recipient and access profile, eligible sources, report revision, reviewer identity, approval timestamp, receipt acknowledgement and audit history.

States are explicit

RelayMap tracks three different kinds of status. A decision can be proposed, approved, rejected or replaced by a later decision. Its evidence can be supportive, contradictory, old, incomplete or unavailable. Its review state can be a candidate, awaiting review, accepted or rejected. These labels must not be collapsed into one: a proposal may be accurately cited while still lacking approval.

Temporal and authority rules

For each record, the data model keeps the date of the underlying event, the source's most recent update and the date RelayMap collected it. A workspace can define which system is trusted for a particular category: for example, an issue tracker may govern task status, while an approved decision register may govern architecture choices. A newer message cannot silently replace a signed-off decision. Where no configured authority resolves a disagreement, both versions are shown for an authorised person to decide.

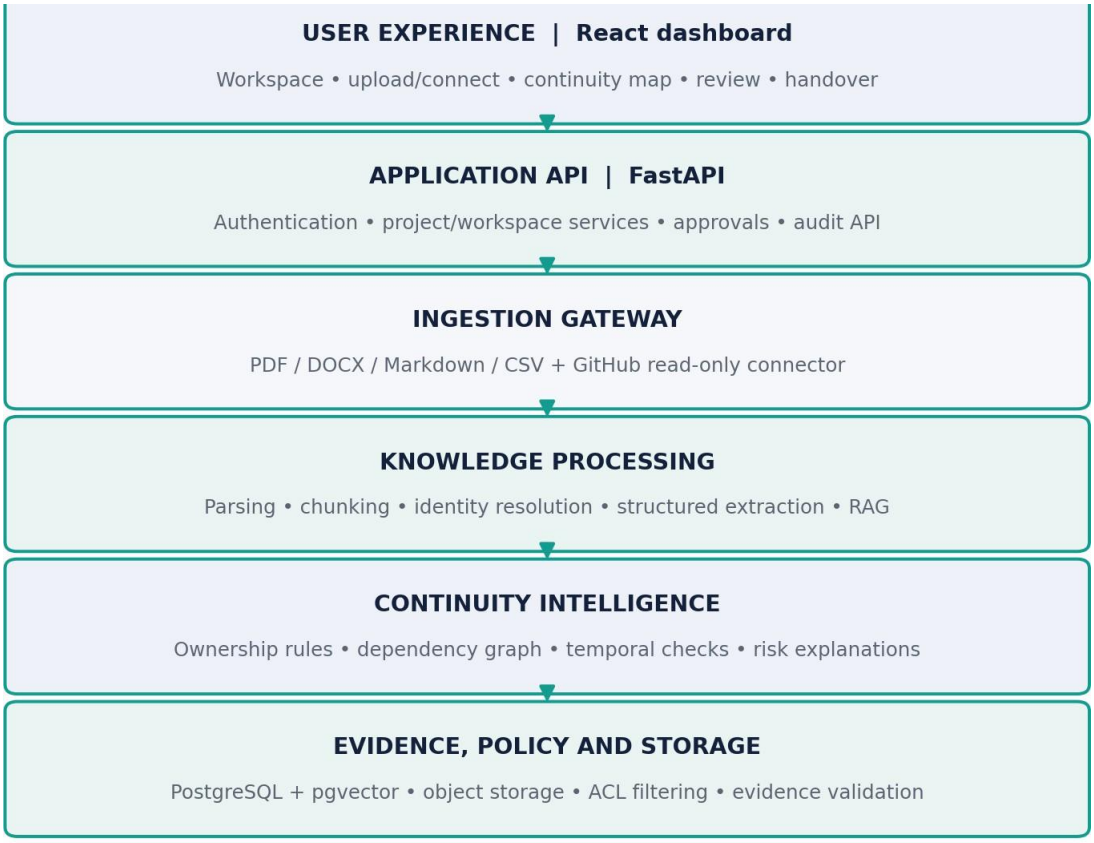
CLAIM-LEVEL EVIDENCE

For each significant statement, RelayMap must provide an accessible source and location or label the statement as an inference, a contributor-supplied note, unsupported or awaiting review. A link that opens successfully is not, by itself, evidence that the referenced passage proves the statement.

06 / TECHNOLOGY STACK AND TECHNICAL DESIGN

System architecture and technology

The design uses familiar components and keeps the first implementation small; neither a dedicated graph database nor a multi-agent framework is required.



Cross-cutting controls: least privilege • source provenance • encryption • secret scanning • audit trail

Layer	Technology / responsibility
Frontend	React.js interface for creating workspaces, checking connections, exploring project relationships, reviewing records and opening handovers.
Backend	FastAPI services in Python for sign-in, project endpoints, workflow coordination, permission enforcement and audit recording.
Ingestion	PyMuPDF and python-docx for document extraction, lightweight readers for Markdown and CSV, and a GitHub API connector configured for read-only access.
AI processing	An LLM adapter that can work with a selected provider for schema-based extraction and source-grounded text; RAG supplies context but does not independently define project state.
Data	PostgreSQL stores users, project objects, access rules, tasks, sources and audit entries; pgvector supports semantic lookup; original uploads are kept in object storage.
Visualisation	A graph visualisation in React can draw the people, work items, decisions and dependencies from the relationships held in PostgreSQL.

Why PostgreSQL is sufficient for the first relationship map

For the first release, PostgreSQL can store project links in indexed relationship tables. A separate graph database should be considered only if real usage later shows that the size or complexity of queries requires it.

06 / TECHNOLOGY STACK AND TECHNICAL DESIGN

How information becomes a trusted handover

A defined processing sequence, with checks at each step and clear handling when something goes wrong.

Key processing sequence

1. Confirm the user's identity and workspace, then establish the permissions that apply to this session.
2. Check each file's format, size and readability. Inspect it for likely credentials or hostile content, and isolate inputs that fail the safety checks.
3. Extract readable content while retaining its original location - such as page, heading, spreadsheet row, issue or comment - together with the source dates.
4. Attach consistent metadata to each item: workspace, source, author, source revision, content fingerprint and the access policy inherited from its origin.
5. Ask the model to return information in a defined schema. Store the result as a reviewable candidate, with its evidence attached, rather than accepting generated text as a fact.
6. Match people and project objects against known identifiers. Keep uncertain matches separate until a person confirms that they refer to the same entity.
7. Use relationships already present in the source platforms as the primary links. Save model-proposed connections separately and mark them as awaiting validation.
8. Apply fixed checks to ownership, task state, successor assignment, source age, disagreements, sync coverage and source integrity.
9. First restrict the evidence set to material the recipient may access. Generate from that filtered set, check every cited reference, and prepare a draft for authorised review.
10. Release the handover only when the authorised reviewer has approved the version prepared for that recipient. Save the approval details and the source snapshot used to create it.

Connector behaviour and completeness

For the first release, GitHub access is restricted to repositories chosen by the project lead and to the records needed for a handover: issues, assigned users, labels, open/closed state, pull-request details and relevant discussion. The connector follows API pagination, saves its progress checkpoint and uses repeat-safe updates so running Sync Now again does not multiply the same records. Each run stores the latest source watermark and the last successful completion time. Rate limits and transient errors receive a limited number of retries. If the run does not finish, its result is marked partial and must not be described as the latest complete project view.

Safe failure behaviour

- **Unavailable input:** tell the user that the source could not be reached; never fill its contents in from a guess.
- **File cannot be read:** isolate the file, explain why it failed and use only the other sources that passed processing.
- **Owner cannot be resolved:** flag the assignment for a person to check, keeping uncertain identities separate.
- **Disagreement between decisions:** present each version with its source and approval state, then ask an authorised reviewer to resolve it.
- **Access withdrawn:** exclude the affected source from further retrieval and sharing, then reassess any handover statements that depended on it.

COMPLETENESS RULE

The dashboard lists successful imports, processing errors and each source's most recent sync. Incomplete coverage is shown plainly and is never labelled as a complete view of the project.

06 / TECHNOLOGY STACK AND TECHNICAL DESIGN

Permission-aware by design

Access restrictions apply throughout the information lifecycle - from import and search to AI generation, approval and delivery.

Permission model

Access is denied unless an explicit rule allows it. A person's broad workspace role does not automatically grant access to every item: permission settings on the source and document determine which records are visible. The first build will enforce rules at workspace, connected-source and document/record level. Restricting individual sections within one document is a possible later extension. Any chunk extracted from a file keeps the file's access restrictions.

- **Pre-search access check: records are filtered by the permission service before semantic search or before content is assembled for the language model.**
- **Access inherited by generated claims: when a statement relies on several sources, it receives the combined restrictions of those sources. A reviewer may change its classification only when authorised and with a recorded reason.**
- **Separate recipient reports: create each handover using only the evidence available to that recipient. Do not create an unrestricted master report and try to secure it by hiding selected paragraphs in the interface.**
- **Protected references: show a source name, excerpt or direct link only when the recipient has permission to open that source.**
- **Permission withdrawal: check access again when a report is generated and when it is opened. If a right has been removed, block the affected view and send dependent handovers back for review.**

Security controls

Threat / failure	Proposed control
Prompt injection in documents	Handle uploaded and retrieved text as data only. It must not override application instructions, alter access policy or initiate tool actions.
Credential leakage	Look for common key, password and token patterns; isolate or remove detected values before model use, and exclude secrets from all handover outputs.
Over-permissioned connector	Grant access only to selected repositories and only for reading; protect connector credentials and allow them to be revoked or replaced.
Unauthorised disclosure	Deny access unless granted, filter before search, carry source restrictions into generated claims and test users with different access levels.
Data exposure to model provider	Limit the text sent to the model to the minimum authorised excerpts. Before deployment, review the provider's available no-training settings and contractual assurances, data retention period and processing region, and record the decision.
Accidental or malicious edits	Use reviewer sign-off, maintain previous revisions and change logs, keep recoverable backups and separate permissions for submitting and approving changes.

No particular model vendor is assumed, and this proposal does not promise privacy protections that have not been verified. Before any non-synthetic project information is sent to a selected provider, the team must review its data retention, model-training, processing-region and security commitments.

06 / TECHNOLOGY AND TECHNICAL DESIGN

A complete handover journey

The following example shows one possible transition in a student software team.

Before the transition

Before the contributor departs, the project lead sets up a workspace and adds both the outgoing contributor and the person taking over. Their platform accounts are verified and the approved sources are connected. Architecture notes and a task export are uploaded, while the GitHub connection is granted read-only access to the selected repository. The lead assigns an access policy to each source. The dashboard then reports what material is available and when each connection last completed a sync.

Reconstruction and gap detection

The analysis identifies an API integration issue assigned to the person who is leaving. The latest completed GitHub sync still shows it as open, and no replacement owner is recorded. Two other tasks are linked to that API work, so the item is raised as a possible ownership gap. Elsewhere, a meeting record suggests changing the authentication design, but the approved decision register still contains the earlier design. RelayMap displays this disagreement and leaves resolution to the authorised reviewer.

Capture what the files do not say

Using the tasks already linked to the contributor, RelayMap prepares a focused set of departure questions. It asks about API limitations, manual environment setup, tests that still need attention, operational contacts and informal workarounds. Answers are stored with the contributor's identity and the date they were provided. They are labelled as statements supplied by that member; they do not become independently verified facts merely because they were captured in the interview.

Review and recipient view

The lead checks the possible ownership gap, records who will take over, and decides how the authentication disagreement will be resolved or assigned for follow-up. Any restricted note is removed from this recipient's version before the report is approved. The successor's handover is generated only from sources they may access; a finance-restricted record is never fetched for inclusion. The incoming developer receives the project snapshot, relevant citations, assigned work, dependencies, open questions and an ordered set of next steps, and can record that the handover was received.

EXAMPLE HANDOVER OUTPUT

Project position: the API integration issue is still open. Former assignee: outgoing backend contributor. New owner: not recorded in this draft. Related work: frontend authentication and system testing. Supporting records: the GitHub issue and linked task entries. Open question: sources disagree about the authentication design. Suggested sequence: appoint an owner, confirm the approved design, finish the API connection and then continue the dependent tests.

06 / TECHNOLOGY STACK AND TECHNICAL DESIGN

MVP scope and explicit non-goals

Keep the first release to the smallest feature set that can demonstrate a full, reviewable transition from source import to handover.

MVP: included

Must work in the first functional build	Definition of done
Workspace and member setup	Open a workspace, add members, connect confirmed accounts and set each person's access.
Static ingestion	Accept the listed file types and retain references to the original page or spreadsheet row.
GitHub connector	Retrieve selected issue and pull-request details, comments, labels, owners and states through restricted read scopes; provide a manual refresh action.
Structured extraction	Save identified decisions, work items, assignees, impediments, outstanding questions and task links in defined data structures.
Evidence and uncertainty	Each important statement points to its source or carries a clear unresolved/uncertain label; source disagreements remain visible.
Continuity Gap Detector	Identify possible ownership gaps only after checking the departing person's verified account, current item status, successor details and source support.
Human review	Allow reviewers to amend, dismiss, validate, redact and approve entries while preserving earlier versions.
Role-specific handover	Build the report from material the recipient may access, with traceable references and practical follow-up steps.

Explicitly out of scope for the MVP

- Always-on collection from all platforms. The initial GitHub connection will be refreshed manually, with clear feedback about whether the sync completed.
- Connectors for Slack, Microsoft Teams, Discord, Google Drive and several task tools. They can be added later through the same connector pattern.
- Actions that alter external systems without approval, including reassigning work, editing code or sending messages.
- Employee ranking, behavioural monitoring, sentiment assessment or automated decisions about employment and performance.
- Copying or retaining passwords, access tokens, API keys or other confidential credentials as part of the transfer.
- Promising to recover information that was never documented, was not shared with the team or cannot legally or technically be accessed.

FEASIBILITY POSITION

A controlled sample workspace and a single GitHub connection are enough to demonstrate the full workflow. This avoids dependence on corporate integrations, specialised graph infrastructure or agents that take actions on their own.

07 / RESPONSIBILITIES OF EACH TEAM MEMBER

Individual Contributions

The development of RelayMap will be carried out by a three-member team, with each member responsible for a specific area of the application. The team will work collaboratively throughout the development process, ensuring that the individual modules integrate into a unified and functional knowledge-handover platform.

1. Ansh – AI Development and Backend Architecture

Ansh will be responsible for designing the system architecture and implementing the AI-powered knowledge-processing engine. He will work on integrating the Large Language Model (LLM), developing the Retrieval-Augmented Generation (RAG) pipeline and creating mechanisms for extracting structured information such as project decisions, responsibilities, unresolved tasks and dependencies from authorised sources. He will also contribute to backend development using Python and FastAPI, including API design, knowledge extraction services, evidence validation and handover generation.

Primary Responsibilities

- AI/LLM integration and RAG implementation.
- Structured knowledge extraction and classification.
- Evidence-linked response generation and validation.
- Backend API development using FastAPI.

2. Griva – Fronted Development and User Experience

Griva will be responsible for designing and developing the user-facing components of RelayMap. Her primary focus will be creating an intuitive and accessible interface that allows users to interact with the platform without requiring technical expertise. She will work on the project workspace, document-upload interface, Project Knowledge Map visualisation, Continuity Risk Dashboard and handover review interface. She will also implement the user interactions required for reviewing extracted knowledge, identifying potential continuity gaps, editing generated information and approving handover reports. Her contribution will ensure that complex AI-generated information is presented in a clear, organised and understandable manner.

Primary Responsibilities

- Frontend development using React.js.
- UI/UX design and dashboard implementation.
- Project Knowledge Map visualisation.
- Continuity Risk Dashboard and status indicators.
- Handover review, correction and approval interfaces.
- Frontend integration and usability testing.

3. Omkar – Project Lead, Data Engineering , Integrations and Database Management

Ansh will be responsible for the data-processing infrastructure and external source integrations within RelayMap. His primary focus will be developing the document-ingestion pipeline and managing the structured storage of project information. He will work on extracting and processing information from supported file formats such as PDF, DOCX, Markdown and CSV, along with implementing the planned GitHub API integration for retrieving authorised issues, pull requests and related project activity. He will also contribute to PostgreSQL database design, pgvector integration, metadata management, identity mapping and source synchronisation. Additionally, he will assist in implementing access-control mechanisms, testing data consistency and verifying the reliability of information transferred between system modules. Additionally, he will coordinate the technical direction of the project, ensure integration between different modules and oversee the overall implementation strategy.

Primary Responsibilities

- Overall technical planning and architecture.
- Document ingestion and processing pipeline.
- GitHub API integration and source synchronisation.
- PostgreSQL database design and management.
- pgvector integration and document retrieval support.
- Metadata extraction and unified identity mapping.
- Data permissions, validation and integration testing.
- Integration and coordination of the complete system.

08 / RESPONSIBILITIES OF EACH TEAM MEMBER

Team Responsibilities

Although each member has a primary area of responsibility, all three members will participate in requirement analysis, system testing, debugging, documentation and final evaluation. Major architectural decisions will be discussed collectively, and each member will review the integration of their respective modules with the rest of the system. This collaborative approach will help maintain development consistency, reduce integration issues and ensure that the final application fulfils the intended objectives of RelayMap.

08 / DELIVERY PLAN

Round-based development plan

This submission is limited to planning. The team will begin substantive product development during the event's authorised build period.

Round	Focus	Planned outcome
Round 1 - Plan	Interview users and define the first use case; specify the workflow, technical design, data structures, permission model, safeguards and achievable scope.	A supported project proposal, a plan for gathering user evidence and a clearly bounded first release.
Round 2 - Build	Build the import process and GitHub refresh, followed by knowledge extraction, source validation, continuity checks, human review and a recipient-facing report.	A judge can start with permitted project records and follow the complete path to a draft handover whose claims and sources can be inspected.
Round 3 - Finish	Strengthen connector stability, access-control testing, handling of evolving facts, departure interviews, ease of use, change history and technical documentation.	A more dependable handover journey, supported by test results, a working demonstration and a realistic next-stage plan.

Suggested workstream ownership

Workstream	Responsibilities
Product, research and UX	Arrange user conversations; document transition cases; shape the navigation, dashboard, handover structure and usability checks.
Backend, data and security	Implement the API layer and data schema, account mapping, authorisation, safe imports, connector refresh, audit records and retention behaviour.
AI, knowledge and evaluation	Define the structured output format, retrieval flow, evidence verification, relationship logic, ownership-gap rules, risk rationale and evaluation dataset.

The workstreams below describe the skills and tasks the project requires; they do not assign work to unnamed individuals. The team must allocate them to its actual members and accurately document each person's contribution under the event rules.

Implementation sequence for Round 2

1. First establish project workspaces, sign-in, access policies and the metadata model for connected sources.
2. Next implement document import and source references for individual claims. Validate that pages, rows and record locations resolve correctly before expanding the AI pipeline.
3. Connect GitHub with read-only scopes, and implement page-by-page retrieval, saved checkpoints, repeat-safe writes, bounded retries and explicit partial-sync reporting.
4. Introduce the knowledge schema, verified account mapping, task-state reconciliation and dependency links that are supported by source evidence.
5. Build the ownership-gap checks and reviewer queue, then add approval controls and handover generation for each authorised recipient.
6. Test access boundaries, incomplete sources, incorrect model outputs, outdated records and user journeys; record the results and known constraints.

08 / DELIVERY PLAN

Research, evaluation and success measures

This section distinguishes research that the team still needs to carry out from evaluation activities planned for the product.

RESEARCH INTEGRITY NOTE

This proposal has not been supplied with completed interview notes, survey responses or observation records. If the team intends to use primary research as evidence, it must carry out the work and retain the records before submission. The material in this appendix is a research plan only; it must not be reported as a finding.

Primary research plan

Speak with approximately five to ten people from student teams, clubs, research groups or small software teams. Where possible, include participants with experience both giving and receiving a handover. Ask them to describe a specific transition they experienced instead of asking only whether they like the concept.

- When someone left or changed roles in your team, which details were hardest for the next person to find?
- Which tools or locations held the decisions, work items, conversations, source code and environment instructions?
- What did the successor need to ask the previous owner because it was not clear from the available records?
- Are there any project records that should be restricted to particular people or responsibilities?
- Which handover details would most reduce delays, repeated investigation or avoidable rework?

For each interview, note the participant's general role, the type of transition, the systems involved, examples of missing context, the workaround used and the RelayMap capability that may address the issue. Avoid collecting personal details that are not needed. Obtain permission before recording or quoting anyone, and report themes without identifying the participants.

Evaluation dataset and method

Build a test workspace from synthetic or authorised material. Include labelled tasks, decisions, user identities and dependencies, along with examples of conflicting notes, stale records, access-restricted sources and failed or partial GitHub updates. Before testing the system, prepare a ground-truth answer set that people have checked manually. Score RelayMap against that reference rather than judging the output by writing quality alone.

Proposed MVP acceptance targets

Measure	Acceptance target for test scenarios
Source traceability	For each important output, either a reviewer can open the relevant evidence location or the system clearly marks that support is missing or pending.
Permission leakage	No test account receives a restricted record or extract in any scenario covered by the access-control test plan.
Orphan detection	Compare flagged items with a human-labelled task set. Aim for at least 85% precision before surfacing these alerts prominently to users.
Sync integrity	All sync failures and incomplete runs are visible, and no incomplete source is described as fully refreshed.
Approval control	A report cannot be shared with its recipient until the correct reviewer has approved that version.
Usability	A person new to the project can locate the initial tasks and inspect their supporting records without help from the team.

The figures and conditions below are targets for future testing, not achievements already demonstrated. The final report must state the measured results and disclose failures or gaps.

09 / EXPECTED IMPACT OF THE SOLUTION

Expected Impact of RelayMap

RelayMap aims to improve project continuity by ensuring that important knowledge, responsibilities and operational context remain accessible when a team member transitions out of a project. The platform is expected to create value for student development teams, startups, volunteer groups and other organisations where project responsibilities are distributed among multiple individuals.

1. Reduced Knowledge Loss

By collecting information from authorised documents, repositories and project discussions, RelayMap will help preserve important technical decisions, implementation details, unresolved issues and responsibilities that might otherwise disappear when an individual leaves.

2. Faster Project Transitions

Incoming members will receive an organised handover containing the project's current state, previous decisions, unfinished work, dependencies and recommended next actions. This can reduce the time spent searching through scattered documents and contacting previous members to reconstruct missing context.

3. Improved Project Continuity

The Continuity Intelligence Engine will identify potentially orphaned tasks, unassigned responsibilities and dependencies affected by a departing member. This will help project leads recognise areas that require attention before they interrupt ongoing development.

4. Better Decision Making

Evidence-linked knowledge records will allow users to understand the reasoning behind important project decisions and inspect the original supporting information. Uncertain, outdated or contradictory information will be highlighted instead of being presented as verified facts.

5. Improving Accountability and Ownership

By identifying responsibilities and their associated tasks, RelayMap will make it easier for project leads to understand which activities require reassignment and which items still need verification or completion.

6. Secure Knowledge Sharing

Permission-aware retrieval, access controls and human approval will help ensure that incoming members receive relevant project information without exposing confidential or unauthorised material.

7. Reduced Repeated Work

By making existing decisions, previous implementations and unresolved problems easier to discover, RelayMap aims to reduce unnecessary duplication of work and help incoming members build upon the progress already achieved.

Expected Overall Outcome

The expected outcome is a more organised, transparent and reliable transition process in which incoming team members can understand the project's current position, identify their responsibilities, inspect supporting evidence and continue ongoing work with fewer information gaps. RelayMap will measure its effectiveness through indicators such as handover preparation time, completeness of critical task ownership, citation validity, orphaned-work detection accuracy and unauthorised information exposure. The actual improvements will be evaluated through structured testing and user feedback rather than assumed in advance.

09 / EXPECTED IMPACT OF THE SOLUTION

Expected problem cases and the safeguards used to limit their effect

It would be unrealistic to claim that all failure is preventable. RelayMap is designed to expose uncertainty early, constrain what the system can do and make its remaining risks clear.

Risk	Mitigation / safe behaviour
Undocumented knowledge	Ask the departing contributor focused questions and retain their answers as member-supplied context. Do not imply the system can retrieve information no one recorded or shared.
False orphan candidate	Check account identity, unresolved status, successor assignment and source evidence before raising a candidate; keep it labelled as possible until a reviewer decides.
Wrong or duplicate identity	Link platform accounts only after verification. Similarity matches may be proposed but must not merge unclear identities; retain the source history and request human confirmation.
Outdated or contradictory facts	Record the event date, source update date and import date; define which sources govern each fact type; flag old or disagreeing records. A newer entry alone cannot replace a signed-off decision.
Unsupported AI claim	Constrain model output to a known schema, attach evidence to individual claims, validate source references in the backend and retain human review. Mark conclusions that remain inferred or unknown.
Permission leakage through RAG	Apply access rules before retrieval and model prompting, carry source restrictions into derived statements and prepare separate reports for each recipient.
Prompt injection / secret exposure	Keep imported text separate from system instructions, prevent it from triggering tools, detect and mask likely secrets, and limit the first connector to read access.
Partial source sync	Persist API page progress and sync checkpoints, retry temporary failures within limits and record the latest retrieved source time. Show incomplete coverage and prevent an unfinished run from receiving a complete status.
AI provider data handling	Send only the minimum permitted context, assess model-provider terms for training, retention and processing location, and stay with synthetic material until that review is complete.
Reviewer overload	Put consequential or disputed records at the top of the queue, allow routine low-impact items to be reviewed together and log all reviewer actions.
Incorrect next-action recommendation	Treat suggested next steps as guidance. A person with the proper authority must make assignments or change another platform.

Residual limitations

There are limits to what the platform can know. It cannot recreate private or deleted discussions, events that were never recorded, or knowledge held only by someone who cannot be consulted. Account ownership may remain uncertain, third-party APIs may be unavailable or change, and a language model may misunderstand a source. Reviewers are also capable of overlooking errors. RelayMap therefore exposes gaps in source coverage, labels uncertain outputs and leaves consequential decisions with authorised people.

10 / FUTURE SCOPE

Future Scope and Product Expansion

Although the initial version of RelayMap will focus on delivering a complete knowledge-handover workflow using authorised document uploads and a GitHub integration, the long-term objective is to evolve the platform into a comprehensive project continuity and institutional knowledge management system. The following enhancements are proposed for development beyond the hackathon.

1. Expanded Third-Party Integrations

Future versions of RelayMap will support additional integrations with commonly used collaboration and project management platforms, including Slack, Microsoft Teams, Discord, Google Drive, Jira and other task-management tools. These integrations will allow the platform to collect authorised information from a wider range of project activities and reconstruct project context across multiple environments. Synchronisation will be designed to respect the access permissions and data-retention policies of the connected platforms.

2. Advanced Knowledge Graph and Dependency Analysis

The Project Knowledge Map can be expanded into a more comprehensive knowledge graph capable of representing relationships between team members, technical decisions, tasks, repositories, dependencies and project milestones. Advanced graph analysis will help identify indirect dependencies, recurring issues and components that may become difficult to maintain after a member leaves.

3. AI-Assisted Departure Interviews

A future version will introduce an intelligent departure interview feature that generates personalised questions based on the departing member's responsibilities and previous project activity. This will help capture practical knowledge, undocumented procedures, recurring problems and important context that may not exist in written records. The captured responses will be labelled by their origin and incorporated into the handover process for review and verification.

4. Advanced Continuity Risk Prediction

RelayMap will be enhanced with more detailed continuity-risk analysis that considers task ownership, dependency relationships, unresolved issues, historical activity and the availability of replacement owners. This will provide project leads with clearer explanations of potential disruptions and help them prioritise the responsibilities requiring immediate attention. Risk indicators will remain explainable and will not be presented as guaranteed predictions of project failure.

5. Advanced Continuity Risk Prediction

Future development may introduce an interactive AI assistant that allows incoming members to ask questions about the inherited project. The assistant will retrieve answers from approved, permission-eligible project knowledge and provide supporting source references. It will also identify questions that cannot be answered reliably because the necessary information is missing or requires confirmation.

6. Advanced Security and Organisational Controls

For adoption by larger organisations, RelayMap can introduce enterprise authentication, single sign-on, more granular document-level permissions, advanced audit logs, configurable data-retention policies, stronger encryption controls and organisation-specific access management. These improvements will support deployments involving multiple teams and confidential organisational information.

7. Handover Lifecycle Management

Beyond generating the handover document, RelayMap can evolve to manage the entire transition process. Future features may include approval notifications, handover acknowledgements, successor assignment, task reassignment workflows, version tracking and confirmation that incoming members have reviewed important responsibilities.

8. Analytics and Performance Monitoring

The platform may introduce organisational dashboards that help project leads understand recurring knowledge gaps, frequently unassigned responsibilities, common transition risks and handover quality across multiple projects. These insights can help organisations improve their documentation practices and prepare for future team transitions.

9. Broader Industry Adoption

While the initial implementation will target student project teams and small development groups, the platform has potential applications in software companies, startups, research laboratories, educational institutions, volunteer organisations and other environments where continuity of knowledge is important. Additional domain-specific templates and workflows can be introduced to support different types of projects.

Long-Term Vision

The long-term vision of RelayMap is to become a reliable knowledge-continuity platform that allows teams to preserve, organise, verify and transfer their collective project knowledge. Rather than functioning only when a member leaves, the platform will continuously help teams maintain an organised record of decisions, responsibilities and project context, making future transitions more predictable and less disruptive.

11 / COMPLETE TEAM AND PARTICIPANT DETAILS

Complete Team and Participant Details

Team Name: omkarpsawardekar47
Team Size: 3 Members
Project Name: RelayMap – Evidence-First Project Continuity and Knowledge Handover Platform
Project Domain: Artificial Intelligence, Data Science and Knowledge Management
Proposed Technology Stack: React.js, Python FastAPI, PostgreSQL, pgvector, LLM API and GitHub API.

Participant 1
Name: Omkar Sawardekar
Email: omkarpsawardekar47@gmail.com
Contact: 9545792989
Institution: AISSMS IOIT
Department: TY AI&DS

Participant 2
Name: Griva Shah
Email: grivashah.2006@gmail.com
Contact: 9403350825
Institution: AISSMS IOIT
Department: TY AI&DS

Participant 3
Name: Ansh Pharate
Email: anshpharate0106ul@gmail.com
Contact: 9156127771
Institution: AISSMS IOIT
Department: TY AI&DS